COBAL

Context-Aware AI Augmentation for More Engaging Offline Racing in Assetto Corsa

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Abstract

This paper presents COBAL (Context-Oriented Behaviour Augmentation Layer), a real-time control prototype implemented in Lua as a proof of concept for augmenting Assetto Corsa's offline AI. COBAL is designed to increase the player's engagement in offline racing by promoting closer and more sustained competition with AI opponents. COBAL is ready to use on supported cars and circuits, with no manual tuning required. Using the per-car AI setters exposed by the Lua software development kit (SDK) of the Custom Shaders Patch (CSP), COBAL continuously schedules selected behavioural controls for each AI pilot according to that pilot's evolving racing context, rather than leaving their values fixed throughout the session. COBAL's shared scenario engine infers this context every frame and encodes it as a seven-component fuzzy membership vector. Live control modules translate this shared context and the race's relative geometry into bounded control demands. The first module, a zero-order Takagi–Sugeno controller, augments each AI pilot's driving commitment, constantly modulating how hard it pushes and how much racing room it concedes. The second module tightens the field and extends its battles through a control law drawn from flocking and consensus control. The third module augments each AI car's use of the racing line: when the player attacks, the covering move escalates with the imminence of the threat, up to the full usable width of the defended side; in clear air, the line carries a subtle, bounded variation, so the field reads as alive. The P2/P3 evaluation compared candidate COBAL parameter sets across 457 six-lap races at Barcelona, principally using fields of eleven identically configured BMW Z4 GT3 opponents. In the leading cohesion configuration of the isolation campaign, the median aggregate AI-pair duel time increased from 136.3 to 374.7 pair-seconds per race, approximately 2.75 times the baseline. Across that campaign's leading configurations, median completed overtakes increased by up to 83 %. The measured gains demonstrate that context-sensitive control and empirical calibration can improve multiple aspects of offline AI racing in Assetto Corsa, producing closer fields, more frequent overtaking and more sustained AI-versus-AI contests under the tested conditions. This proof of concept marks the beginning of a broader development programme, with further behavioural features and expanded car and circuit support planned. Its modular architecture provides a foundation for exploring new forms of augmentation beyond the present feature set, with user feedback guiding refinement and future directions.

1 Introduction

Wheel-to-wheel racing continually places a driver in distinct situations: a start, an attack, a defence, a closing gap. A human driver's caution, aggression, braking commitment, and racing line vary continuously with these situations.

Assetto Corsa's built-in AI handles steering, braking, acceleration and interaction with other cars [1]. Custom Shaders Patch (CSP), a community-developed extension led by Ilja Jusupov, expands the game with additional features, including revised AI behaviour and a Lua software development kit (SDK) [2, 3]. CSP provides the Lua environment in which COBAL runs.

COBAL augments the native AI to make offline races more competitive throughout the field. Opponents launch more consistently, remain within racing reach, pursue and defend positions with greater commitment, and create more opportunities for sustained battles and overtakes. In eligible player-facing encounters they respond to surrounding traffic and available track space; restrained line variation and calmer recovery give those encounters continuity. Assetto Corsa's AI drives each car while COBAL adapts its behaviour as the race unfolds. The driver selects a supported race, then races without configuring the behavioural modules.

Modern racing titles pursue several approaches to convincing opponents. rFactor 2 combines per-driver talent settings with optional learning of track lines [4]; Le Mans Ultimate continues development of AI racecraft in that lineage [5]. Automobilista 2 characterises drivers through personality files spanning skill, aggression, defending, consistency and susceptibility to errors [6]. Published work by Nic Melder, then at Codemasters, and Simon Tomlinson describes layered behavioural control and runtime race-management logic [7, 8]. Gran Turismo Sophy demonstrates another approach: a learned driving policy trained through deep reinforcement learning to compete with world-class drivers [9].

Within Assetto Corsa, Custom Shaders Patch extends the native AI and provides a platform for community development [10]. COBAL combines a shared fuzzy description of each car's racing situation, coordinated behavioural augmentation and controlled race-level evaluation to improve offline competition.

Racing-AI research has achieved superhuman pace and extended learned policies to vision-based driving [9, 11, 12]. Player-experience research brings a complementary perspective. In a controlled comparison, players preferred the slower, more variable opponent and rated it more realistic [13]. A large online-racing quasi-experiment found that AI opponents' effect on engagement varied with competition intensity, immersion, motivation and skill [14]. COBAL's objective follows this player-experience perspective: engagement, understood as closer packs, more wheel-to-wheel racing, and more contested positions. The Calibration Measurement Campaign measures these competitive conditions.

This paper presents three COBAL contributions: a shared engine that recognises each car's racing situation across circuits and coordinates in-race augmentation without behavioural setup by the player; a staged calibration methodology combining racing-activity metrics, race-level estimation, physical-spacing checks and circuit-specific safety analysis; and measured BMW-field outcomes in isolated and combined configurations.

The experiments reported here principally concern BMW Z4 GT3 racing at Barcelona GP. COBAL supports eleven GT3 cars at that circuit through car-specific calibration.

2 System architecture

COBAL is a modular, context-aware control architecture for real-time augmentation of Assetto Corsa's offline AI. It separates shared perception, behavioural control and bounded actuation. The Scenario Context Engine (SCE) provides the common situational description on which the augmentation modules operate; their control demands are applied through the Lua software development kit of Custom Shaders Patch, while Assetto Corsa's native AI performs the driving [3].

SCE continually interprets each car's racing context from live relative motion and track geometry. Its shared fuzzy representation allows concurrent situations to inform complementary responses: Driving Commitment Augmentation (DCA) governs competitive commitment, the Car Cohesion Modulator (CCM) governs field cohesion, and the Cooperative Positioning Module (CPM) governs lateral positioning in eligible player-facing encounters. Perception and

augmentation evaluation run at simulation update cadence, so these responses follow the development of each encounter.

Control responsibilities are partitioned by module and race phase, with a single writer assigned to each governed control. This establishes structural orthogonality: modules retain distinct control authority and can be calibrated individually, while their behavioural interactions are evaluated in combination. Race Start Augmentation (RSA) governs the launch phase before releasing its controls for subsequent augmentation.

The architecture is conceived for empirical calibration: individual-module measurements establish candidate settings, and combined-system and circuit-specific evaluation assess their interaction and driving effects. On supported content, COBAL applies the packaged calibration and situational adaptation automatically, without requiring behavioural configuration or controller tuning from the player. Shared perception and explicit control ownership also provide a foundation for further behavioural features and expanded car and circuit support.

2.1 Scenario Context Engine

At each simulation update, SCE assesses what is happening around every car, including the player. It draws on live gaps, closing motion, relative positions and available track room to identify pursuit, close running, duels, pressure and recovery. COBAL's modules use that changing assessment to adapt opponents during the race. The driver chooses a supported race, then drives; SCE supplies the situational awareness automatically.

SCE is circuit-agnostic because it recognises racing situations through live interactions among cars. It also considers the room available at each car's current position, so a pursuit or duel is understood in its immediate surroundings. The same engine follows encounters around the lap and can serve additional circuits as COBAL expands its calibrated support (Section 3). The player does not have to configure situations for each track.

Grounded in fuzzy set theory [15], SCE assigns each car graded membership in seven situations: normal running, pursuit, close running ahead, duelling, close running behind, pressure and recovery. Each value expresses how strongly a situation applies. The values are not probabilities that must sum to one: a car can pursue one rival while another approaches from behind. This preserves overlapping situations and gradual change as a fight develops.

One SCE lets the commitment, cohesion and positioning modules work from the same reading of the race; positioning also considers the immediate space around the player. Recorded duel activity and the optional leaderboard draw on the same classification, connecting what the player sees to what COBAL measures. Situation-awareness research motivates maintaining this current account of surrounding conditions [16]. Shared context lets COBAL coordinate its responses automatically while the player concentrates on racing.

2.2 Functional components

The behavioural components described here address distinct competitive tasks during the launch phase and the subsequent race. Table 1 connects each component's responsibility to its on-track effect.

Table 1: COBAL components, their behavioural responsibilities and racing effects.

Component	Behavioural responsibility and racing effect
RSA — Race Start Augmentation	Reinforces consistent, competitive acceleration from lights out, making positions on the opening straight more contested.
DCA — Driving Commitment Augmentation	Responds to pursuit, duels and pressure with greater competitive commitment, making close attacks and defences more persistent.
CCM — Car Cohesion Modulator	Keeps more opponents within racing reach for longer, sustaining a closer field and repeated opportunities for duels and overtakes.
CPM — Cooperative Positioning Module	Provides player-facing defence that responds to nearby traffic and usable track space, and adds restrained, smooth line variation from the nearby opponent ahead between battles.
Recovery response	Eases augmentation after an off-track excursion, contact or substantial slowdown, supporting a calmer, more orderly return to racing.

CCM's field-level objective is competitive continuity: keeping opponents within reach creates repeated opportunities for attack, defence and overtaking. Research on multi-agent coordination and group structure provides scientific context for this objective [17, 18]. The experiments measure both physical proximity and the duels and passes it enables.

DCA uses Takagi–Sugeno fuzzy inference to translate SCE's concurrent situation grades into a continuous commitment response [19]. Fuzzy inference has precedents in simulated-car racing [20]; here it lets commitment develop with pursuit, close fighting and pressure rather than remain fixed for an entire session.

CPM addresses the spatial dimension of a contest through geometry-aware positioning and continuity of motion. A direct challenge produces recognisable defensive positioning; between battles, the nearby opponent ahead shows restrained line variation. Circuit review assesses these movements together with available room and surrounding traffic.

Together, CCM, DCA and CPM give the player a field that stays close, contests positions and uses the available space responsively. The evaluation connects that racing experience to passes, sustained duels, physical spacing and driving quality.

COBAL's contextual awareness of closing traffic helps avoid some rear-end contacts, while geometry-aware positioning responds to surrounding cars and usable room. Following a disruption, the coordinated recovery response eases augmentation while the native AI resumes stable racing.

2.3 Player-facing operation

COBAL supports dry racing at Barcelona GP with the eleven GT3 cars across the grid. It requires no tuning or behavioural configuration from the user. SCE assesses every participant's racing situation as the race unfolds.

RSA strengthens the launch; DCA, CCM and CPM sustain the subsequent attacks, defences and close-running opportunities.

The optional *COBAL AI Leaderboard* adds opponents' racing situations to positions, gaps and lap times. It makes developing pursuits, duels, pressure and recovery visible alongside the race itself.

2.4 Empirical implications

COBAL's central racing effects have distinct measurable consequences. SCE's DUEL classification anchors sustained duel time; completed overtakes and duel conversion capture changes of position; physical pack gap tests whether opponents remain within racing reach. Pace and safety observations assess those gains alongside driving quality. Section 3 specifies the controlled comparisons, and Section 4 reports the BMW-field results.

3 Calibration and empirical evaluation

3.1 Purpose and phase sequence

In this paper, *calibration* is the development-time selection, refinement or retention of settings for a defined supported racing environment. *Evaluation* is the measurement and comparison of the resulting racing behaviour. The two activities are linked throughout the Calibration Measurement Campaign (CMC): each phase poses a different decision question and uses observations suited to that question. Measurement and setting selection are integrated throughout the calibration sequence. From P2 onward, the phases share race-level measures of engagement, cohesion, pace and safety, supplemented by circuit diagnosis in P4.

The five phases progress from exploratory response checks to an assessed configuration (Table 2). P0 establishes whether an investigated control has an observable effect and its direction. P1, including a later complementary sweep, characterises the range of that response and supplies empirical constraints for design. Together these phases establish the empirical constraints used for value selection in P2. P2 is the value-producing stage: it examines candidate combinations for DCA and CCM in separate module studies within the campaign. P3 tests the selected candidates together, measuring whether their effects remain useful in the combined system. P4 evaluates driving and racing across the circuit, with particular attention to connected manoeuvres, recovery and safety; it can support a refinement or a decision to retain the existing configuration.

Table 2: CMC phases: the decision question, evidence and use of each phase.

Phase	Decision question	Output and use
P0	Is an investigated control observable in racing, and in which direction does it act?	Response verdict and direction; identifies what merits a sensitivity study.
P1	How does the response change across the investigated range?	Empirical bounds and interaction evidence that guide module design and subsequent value selection.
P2	Which candidate combinations serve each module's objective?	Separate DCA (P2a) and CCM (P2b) calibration against a campaign baseline; candidate settings for combined assessment.
P3	Do selected module candidates remain effective together?	Matched-condition comparisons of combined configurations, their interactions and their contribution to racing outcomes.
P4	How does the resulting configuration drive and race across the circuit?	Whole-race and circuit-specific evidence on driving quality, engagement and safety; supported refinement or retention.

Within P2, DCA and CCM candidates are assessed separately with the other module neutral; interacting choices within a module are assessed together. P3 tests the selected combination under matched conditions. P4 relates whole-race outcomes to circuit passages. P0–P1 use focused

response measures; P2–P4 use common race-level outcomes to answer distinct configuration questions.

3.2 Outcome definitions

Racing activity is measured by duel time, completed overtakes and duel conversion. Physical pack gap measures cohesion. These outcomes describe different aspects of competitive racing.

Duel time measures duel episodes between pairs of AI cars, reconstructed from the recorded DUEL state. Either car can identify a duel, and overlapping intervals for the same pair count once. Interruptions of up to 0.2 s are bridged. Duel time is measured in pair-seconds per race; simultaneous duels between different pairs contribute separately. Completed overtakes count confirmed on-track AI–AI passes, including re-passes. Duel conversion is the share of duelling pairs that also record a pass.

Physical spacing establishes whether small time gaps also correspond to small distances between cars. Let g_i be the forward distance along the track in metres for valid matched AI sample i, estimated from recorded spline positions. Retain samples whose distance is positive and whose recorded time-gap estimate to the nearest car ahead at the same sample is at most 3.0 s. The median in-pack distance over those retained samples is

$$G = \mathrm{median}_i \, g_i. \tag{1}$$

The fraction of matched samples whose time gap exceeds 3.0 s tracks cars outside the pack. Time spent within 1.0, 0.6 and 0.25 s of the car ahead provides additional proximity measures. Pack spacing is reported in metres rather than inferred from time gaps alone.

3.3 Campaign design and controlled conditions

P2 and the active P3 configurations were tested at Barcelona GP with eleven BMW Z4 GT3 AI cars in six-lap races. Only the AI field was scored; the player's car remained parked. The P2 cohesion campaign tested 60 active configurations and an in-campaign baseline. The reported P2 results cover 301 races across 61 configurations, with four or five races per configuration. The P3 campaign tested three module combinations at each of five selected settings, with ten BMW races per active configuration (150 BMW races). The experiments isolated in-race augmentation with RSA inactive, and telemetry began after the warm-up. Six additional baseline races used a different grid and feature state and remain in the campaign record.

Controlled operating point. The P2 and active P3 configurations shared 20 % native aggression and verified Slick Medium tyres (index 1). Conditions were clear, with ambient/road temperatures of 26/37 °C and track state fixed at 98 %. Wind, track evolution (including its randomness), tyre wear, fuel consumption, damage and penalties were disabled.

These AI-field measurements guided subsequent player-focused positioning for the current candidate.

3.4 Circuit-specific driving and safety assessment

P4 examines braking onset, cornering speed and line, exit acceleration and incident sequences, together with engagement and pace over the whole race. Its safety objective is to reduce accident-related noncompletion relative to comparable native AI while sustaining competitive racing. To identify how an incident develops, the analysis follows each car and the surrounding traffic through the approach, corner and exit.

For each configuration and accepted race, the safety measure is the number of confirmed accident-related AI retirements divided by the number of AI starters with an observed outcome. This fraction is calculated only when at least one starter has an observed outcome. Comparisons use independent races and retain the underlying counts. Corner passages and telemetry samples help diagnose incidents; uncertain outcomes are classified separately. Safety and racing activity are evaluated together to preserve the competitive experience.

3.5 Race-level analysis and comparison

Each independent race provides measures of pace, engagement, physical cohesion and safety. Each outcome is summarised by its median across the candidate's accepted races. A longer battle, an additional pass and a smaller physical gap represent different improvements in racing, so these outcomes are reported separately. They are assessed together with pace and safety to judge whether a candidate meets its module's objective and improves the race as a whole.

Configurations are compared by the difference between their medians for the same outcome. P2 compares each candidate with its campaign baseline. The P3 results reported here compare active BMW configurations under matching race conditions. P4 adds comparisons of whole races and individual circuit sections. In each phase, these differences inform the corresponding calibration decision.

The race is the experimental unit; cars and telemetry frames within a race are dependent observations. For configuration outcomes $x_1, \ldots, x_n$, the median and normalised median absolute deviation (MAD) describe performance and variation:

$$m = \operatorname{median}_i x_i, \qquad \mathrm{MAD} = 1.4826 \operatorname{median}_i |x_i - m|. \tag{2}$$

These robust summaries limit the influence of extreme races [21]. Effect sizes describe the magnitude and consistency of differences between configurations.

For any one outcome, let $x_1, \ldots, x_{n_x}$ and $y_1, \ldots, y_{n_y}$ be race outcomes from the tested and comparison configurations. Comparing every race outcome in one group with every outcome in the other gives $n_x n_y$ comparisons. Let F count comparisons favourable to the tested configuration and T count ties. Higher duel time or overtake counts, and lower pack gaps, are favourable. The probability-of-superiority estimate is [22]

$$\widehat{A} = \frac{F + \frac{1}{2}T}{n_x n_y}. \tag{3}$$

A value of $\widehat{A} = 0.5$ means favourable and unfavourable comparisons are equally frequent; 1 means every comparison favours the tested configuration. The Hodges–Lehmann estimate expresses the difference in the outcome's original units [23]:

$$\widehat{\Delta}^{\mathrm{HL}} = \operatorname{median}_{i,j}(x_i - y_j). \tag{4}$$

This is the median of all differences between a tested-group outcome and a comparison-group outcome. It need not equal the difference between the group medians. Negative values favour tighter pack gaps; positive values favour more duelling and overtakes. Rank-sum tests [24] complement these descriptive estimates. Ordered-trend tests [25, 26] assess sweeps of ordered settings, while P3 compares module combinations.

3.6 Comparison design

P2 uses its campaign baseline. P3 compares active BMW configurations with matched grids and feature settings. Environment, tyre and session settings are controlled. Fixed rules define

how outcomes are scored and races excluded, and recorded configuration flags and telemetry confirm the tested conditions.

Reporting duel time, overtakes and physical pack gap together shows how racing activity relates to the spacing between cars.

4 Results

The BMW isolation campaign shows gains in sustained duelling, overtaking and physical cohesion. Results are presented by phase and configuration to identify the conditions in which each effect was measured.

4.1 Isolation-campaign response

In P2 isolation, all 60 active cohesion configurations increased sustained duel time relative to their BMW baseline. The strongest raised its median from 136.3 to 374.7 pair-seconds per race, while completed overtakes rose from 29 to 47 and in-pack spacing decreased from 19.9 to 17.7 m. Across the nine configurations with the highest median duel time (Table 3), median duel time increased by 117–175 %, completed overtakes by 38–83 %, and median in-pack spacing decreased by 9–14 %. The P2 results show gains across complementary measures of close racing.

Table 3: Isolation-campaign baseline and the nine configurations with highest median duel time. The baseline has five races; selected configurations have four or five. Ranges cover per-configuration medians.

Outcome	Baseline	Leading region	Change from baseline
Sustained duel time (pair-s/race)	136.3	295.6–374.7	+117 to +175 %
Completed overtakes (/race)	29	40–53	+38 to +83 %
Duel conversion	0.60	0.67–0.80	+0.07 to +0.20
In-pack spacing (m)	19.9	17.1–18.2	−1.7 to −2.8 m

All nine selected configurations combined longer duel time and more overtakes with tighter physical spacing. These results support COBAL's aim of sustaining battles and creating additional passing opportunities in the tested field.

4.2 Combined-configuration racing

P3 compared three module combinations at each of five selected settings. Their differences in racing activity and spacing guided configuration selection. Table 4 reports one setting, with ten BMW races for each combination. All three combinations were tested under the same race specification.

Table 4: P3 BMW active-arm comparison at one selected setting. Entries are race medians, with ten races per arm.

P3 arm	Duel time (pair-s/race)	Overtakes (/race)	In-pack spacing (m)
CCM	331.4	51.5	17.30
CCM + DCA	285.4	48.5	20.40
CCM + DCA + CPM	297.2	52.5	19.75

At this setting, CCM alone provided the longest duels and tightest spacing, while adding CPM to CCM + DCA increased duel time and completed overtakes and tightened spacing. Testing the modules together guided selection of the player-focused positioning policy.

4.3 Circuit-specific outcome

The final BMW P4 comparison included 57 of 60 scheduled races, with three excluded from analysis. Circuit review refined the player-facing positioning policy while retaining the DCA and CCM settings selected in P3. Tracing incidents distinguished prolonged stopping from other reasons for not completing a race and informed the current policy.

5 Discussion

P2 shows longer battles, more passes and closer physical spacing in the BMW field. P3 tests the modules together, and P4 assesses driving and safety across the circuit. These measured AI-field outcomes guide development of the player-facing racing experience.

In race terms, COBAL begins with more competitive launches, then keeps opponents within reach as they pursue, defend and race alongside one another. Context-sensitive commitment and space-aware positioning sustain meaningful attacks and defences; restrained line variation adds movement between battles, and the recovery response supports a calmer return after disruption. These coordinated behaviours create more opportunities for overtakes and sustained battles visible to the player.

6 Conclusions

In BMW Z4 GT3 racing at Barcelona GP, COBAL's leading isolated cohesion configuration delivered about 2.75 times its baseline duel time, with more overtakes and tighter physical spacing. Comparisons of module combinations quantified their effects on racing, while circuit review guided selection of the driving and positioning policies. Calibration proceeds in stages, linking measured racing gains to configuration and safety decisions.

COBAL augments Assetto Corsa's offline racing with more competitive, responsive opponents and requires no tuning or behavioural configuration from the user. Stronger launches, situational commitment, field cohesion, player-facing positioning and composed recovery sustain the challenge across the race; the optional leaderboard makes opponents' situations visible. Assetto Corsa's native AI drives the cars while COBAL continually augments how they compete.

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